

02 Exploratory Data Analysis

EP

2026-02-20

Analysis of the Grid Model Data

Following creation of the final grid, exploratory data analysis is carried out to determine the types of models to be used and illustrate the data.

```
library(targets)
library(sf)
library(dplyr)
library(tidyr)
library(purrr)
library(ggplot2)
library(patchwork)
library(GGally)
library(skimr)
```

#If running with downloaded grid data use:

```
grid_model_all <- st_read("Data/grid_model.gpkg", quiet = TRUE)

df <- grid_model_all %>%
  sf::st_drop_geometry()
```

#If running after running the pipeline use:

```
#setwd("~/CSCT_Pipeline")
#grid_model_all <- tar_read(grid_model_all)

#df <- grid_model_all %>%
#  sf::st_drop_geometry()
```

```
predictors <- c("n_days", "dtm_elev", "avg_imperv", "rofsw_risk", "rofrs_risk",
               "avg_area_ha", "treeht", "weighted_precip", "weighted_temp",
               "dist_to_river", "pct_greenpace")
```

Plot the Study Area for each city

The following plots show each cities boundary (in red) and the 1km² grid squares (black).

```

#setwd("~/CSCT_Pipeline") Uncomment if working from pipeline, otherwise use downloaded files from GitHub
project_crs <- 27700

# Create cities table
cities <- tibble::tibble(
  city_label = c("Bristol", "Gateshead", "Newcastle", "Exeter", "Nottingham"),
  city_name = c(
    "City of Bristol (B)",
    "Gateshead District (B)",
    "Newcastle upon Tyne District (B)",
    "Exeter District (B)",
    "City of Nottingham (B)"
  ),
  grid_file = c(
    "Data/grid_1km_Bristol.gpkg",
    "Data/grid_1km_Gateshead.gpkg",
    "Data/grid_1km_Newcastle.gpkg",
    "Data/grid_1km_Exeter.gpkg",
    "Data/grid_1km_Nottingham.gpkg"
  )
)

# Read spatial grid (looped) and transform CRS
grids <- cities %>%
  mutate(grid = map(grid_file, ~ st_read(.x, quiet = TRUE) %>% st_transform(project_crs)))

# Add city names and combine grids
grid_model_all <- map2(grids$grid, grids$city_label, ~ mutate(.x, city_label = .y)) %>%
  bind_rows()

# Read boundaries file and filter for my cities
boundary <- st_read("Data/OS_borders/district_borough_unitary_region.shp", quiet = TRUE) %>%
  st_transform(project_crs)

boundaries <- boundary %>%
  inner_join(cities %>% select(city_name, city_label), by = c("NAME" = "city_name"))

# Which grid is in which boundary?
grid_cities <- st_within(grid_model_all, boundaries %>% select(city_label))

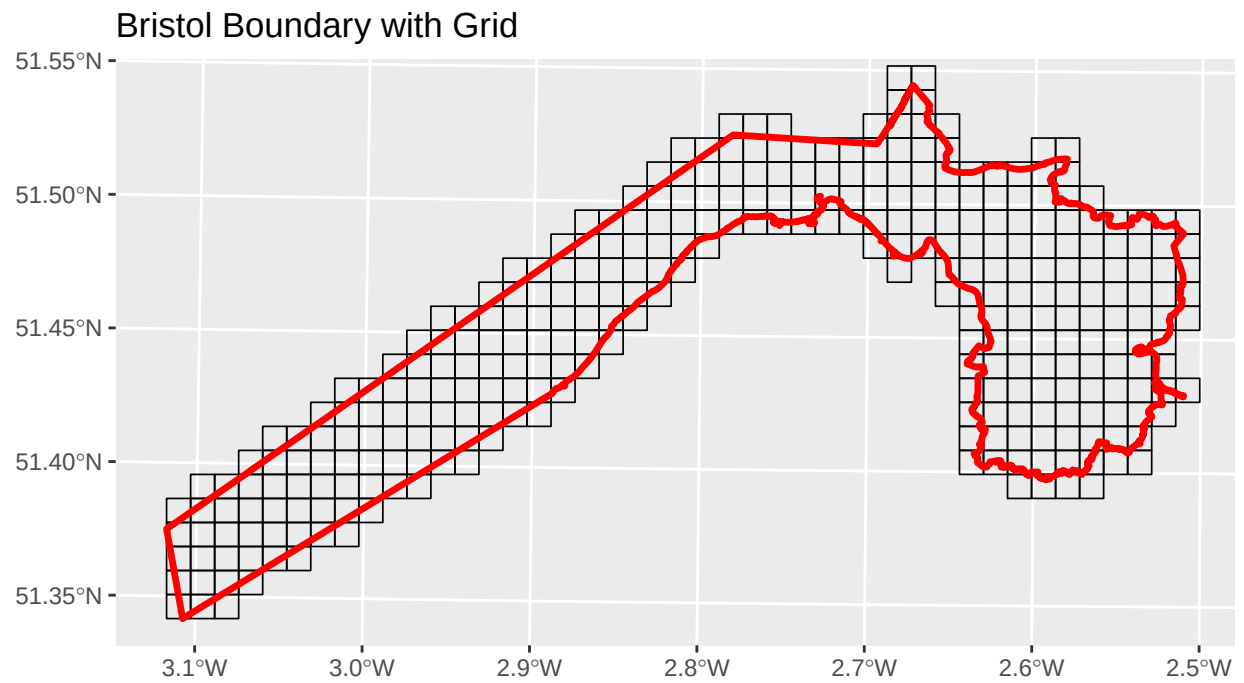
# Plot function
plot_grid <- function(city){
  city_grid <- grid_model_all %>% filter(city_label == city)
  city_boundary <- boundaries %>% filter(city_label == city)

  ggplot() +
    geom_sf(data = city_grid, colour = "black", fill = NA, linewidth = 0.3) +
    geom_sf(data = city_boundary, colour = "red", fill = NA, linewidth = 1.2) +
    coord_sf(
      xlim = st_bbox(city_boundary)[c("xmin", "xmax")],
      ylim = st_bbox(city_boundary)[c("ymin", "ymax")],
      expand = TRUE) +
    labs(title = paste(city, "Boundary with Grid"), x = NULL, y = NULL) +
    theme_gray()
}

plots <- set_names(cities$city_label) %>%

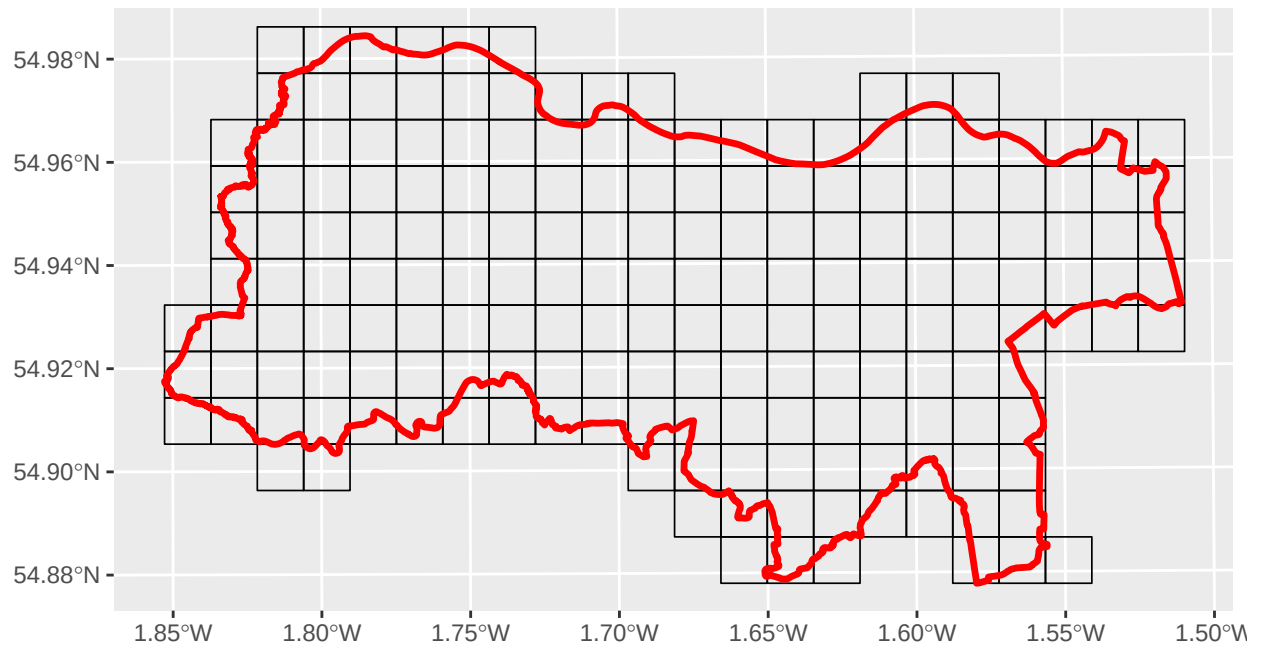
```

```
map(plot_grid)
plots$Bristol
```



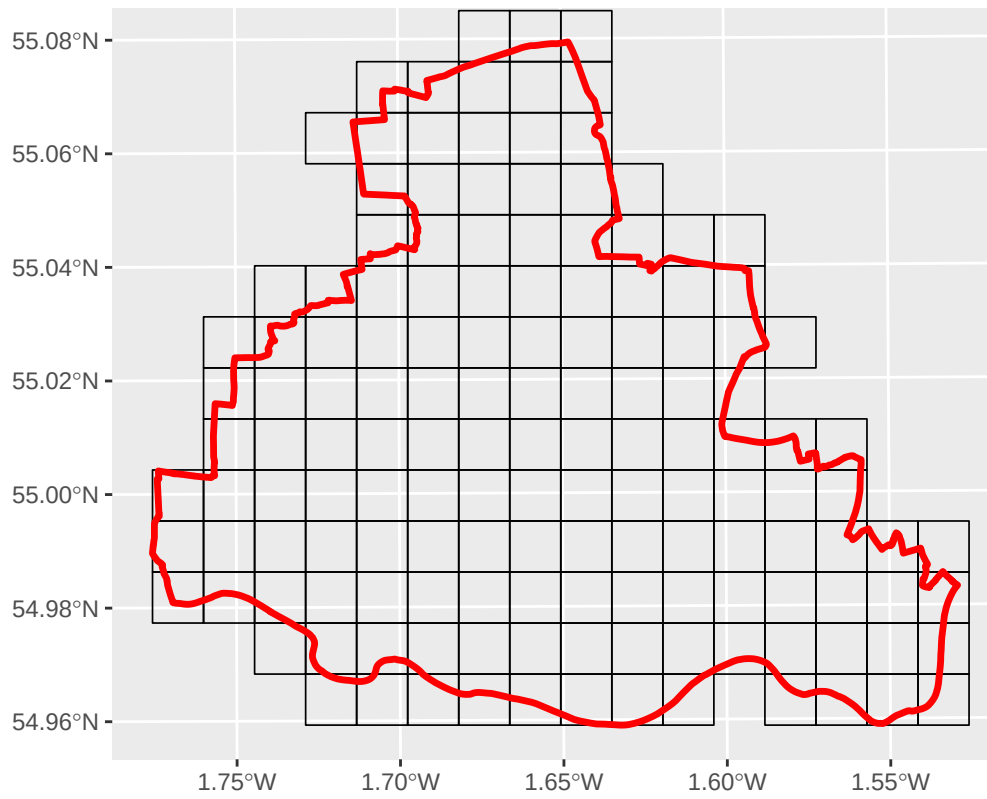
```
plots$Gateshead
```

Gateshead Boundary with Grid



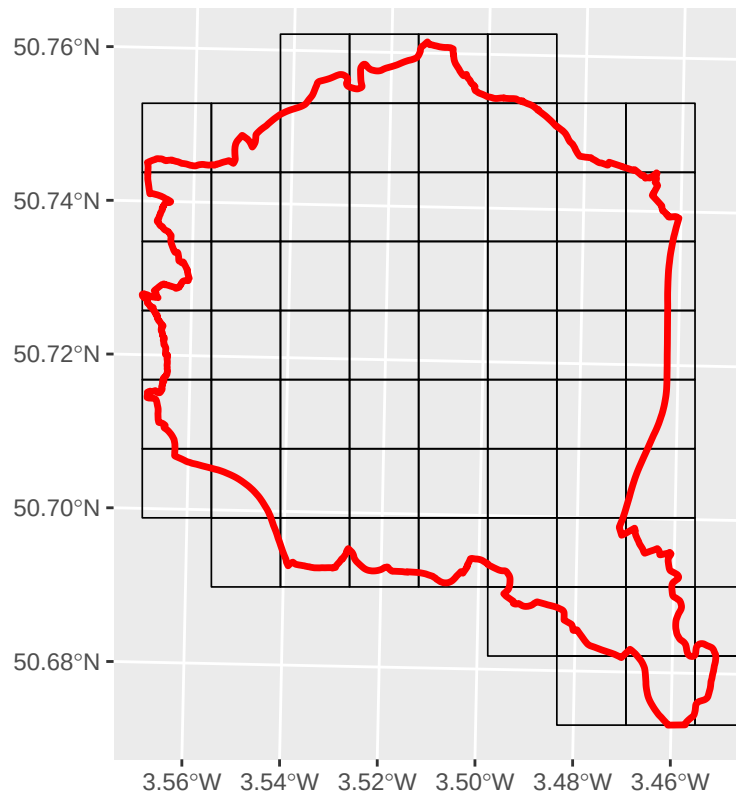
plots\$Newcastle

Newcastle Boundary with Grid



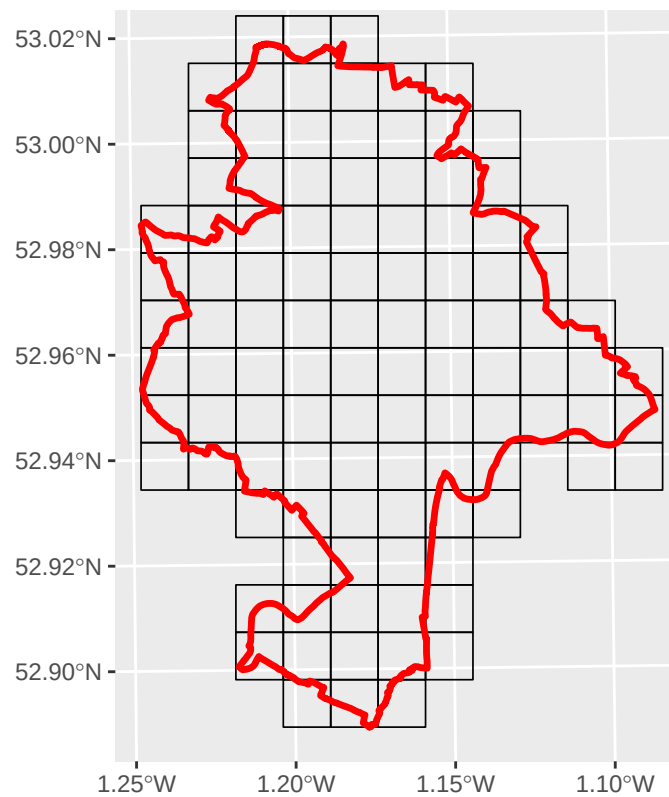
plots\$Exeter

Exeter Boundary with Grid



plots\$Nottingham

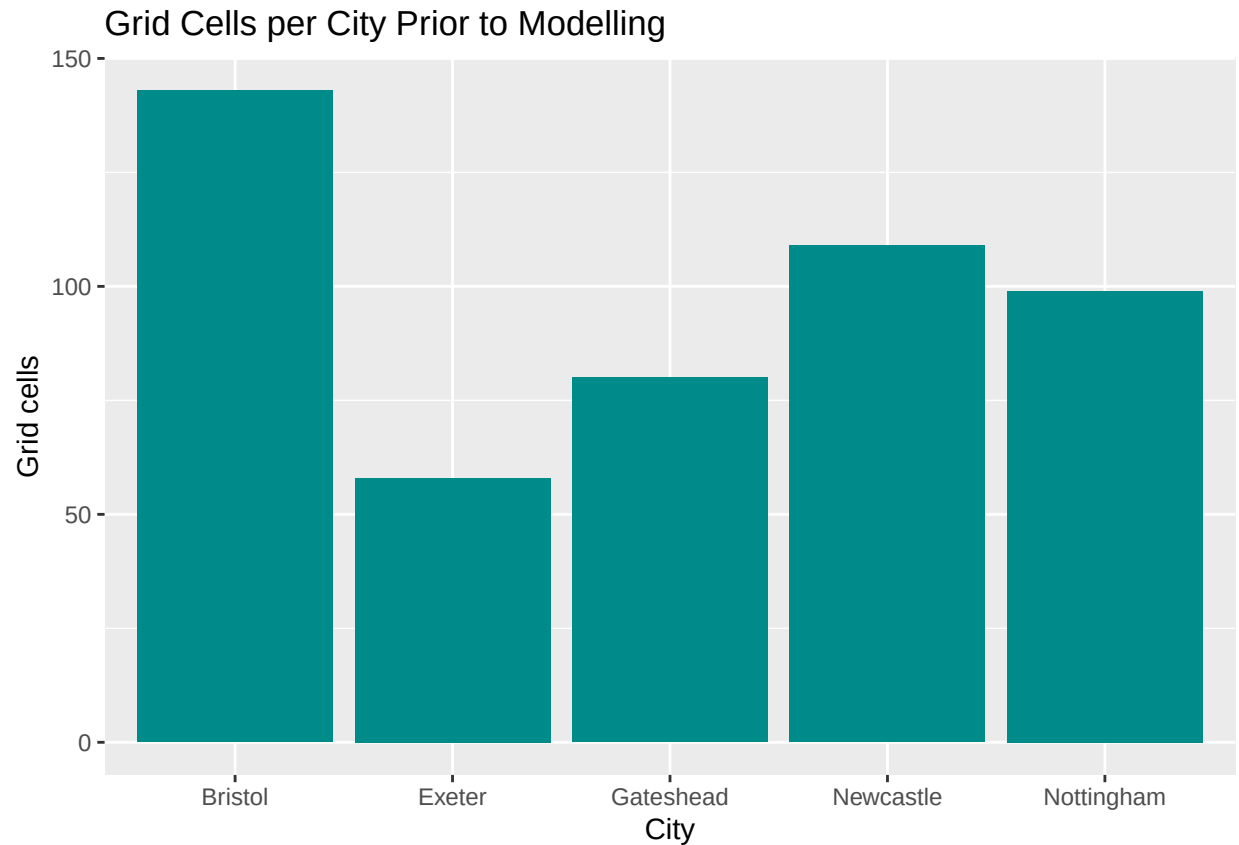
Nottingham Boundary with Grid



Number of grid cells

These plots show the number of grid cells per city, prior to modelling. This is the total number of grid squares, before rows are removed due to missingness.

```
cityplot<- df %>%  
  count(city) %>%  
  ggplot(aes(city, n)) +  
  geom_col(fill = "cyan4") +  
  labs(x = "City", y = "Grid cells", title = "Grid Cells per City Prior to Modelling") +  
  theme_grey()  
cityplot
```



Numerical Summary Statistics

These summary statistics look at the numerical variables in the grid.

```
numerical_predictors <- c("n_days", "dtm_elev", "avg_imperv",  
                          "avg_area_ha", "treeht", "weighted_precip", "weighted_temp",  
                          "dist_to_river", "pct_greenpace")
```

```
summary_table <- df %>%  
  select(all_of(numerical_predictors)) %>%  
  pivot_longer(  
    cols = everything(),  
    names_to = "Predictor",  
    values_to = "Value") %>%  
  group_by(Predictor) %>%  
  summarise(  
    Mean = mean(Value, na.rm = TRUE),  
    SD = sd(Value, na.rm = TRUE),  
    Min = min(Value, na.rm = TRUE),  
    Max = max(Value, na.rm = TRUE),  
    .groups = "drop")
```

```
summary_table
```

```
## # A tibble: 9 x 5
```


##	Predictor	Mean	SD	Min	Max
##	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	avg_area_ha	4.80	10.8	0	181.
## 2	avg_imperv	36.2	19.1	0.435	84.3
## 3	dist_to_river	718.	605.	1.18	2994.
## 4	dtm_elev	51.4	30.9	1.06	152.
## 5	n_days	386.	594.	1	3955
## 6	pct_greenpace	0.126	0.160	0	1.27
## 7	treeht	5.01	1.46	0	10.5
## 8	weighted_precip	771.	95.5	639.	1017.
## 9	weighted_temp	10.3	0.616	9.19	11.1

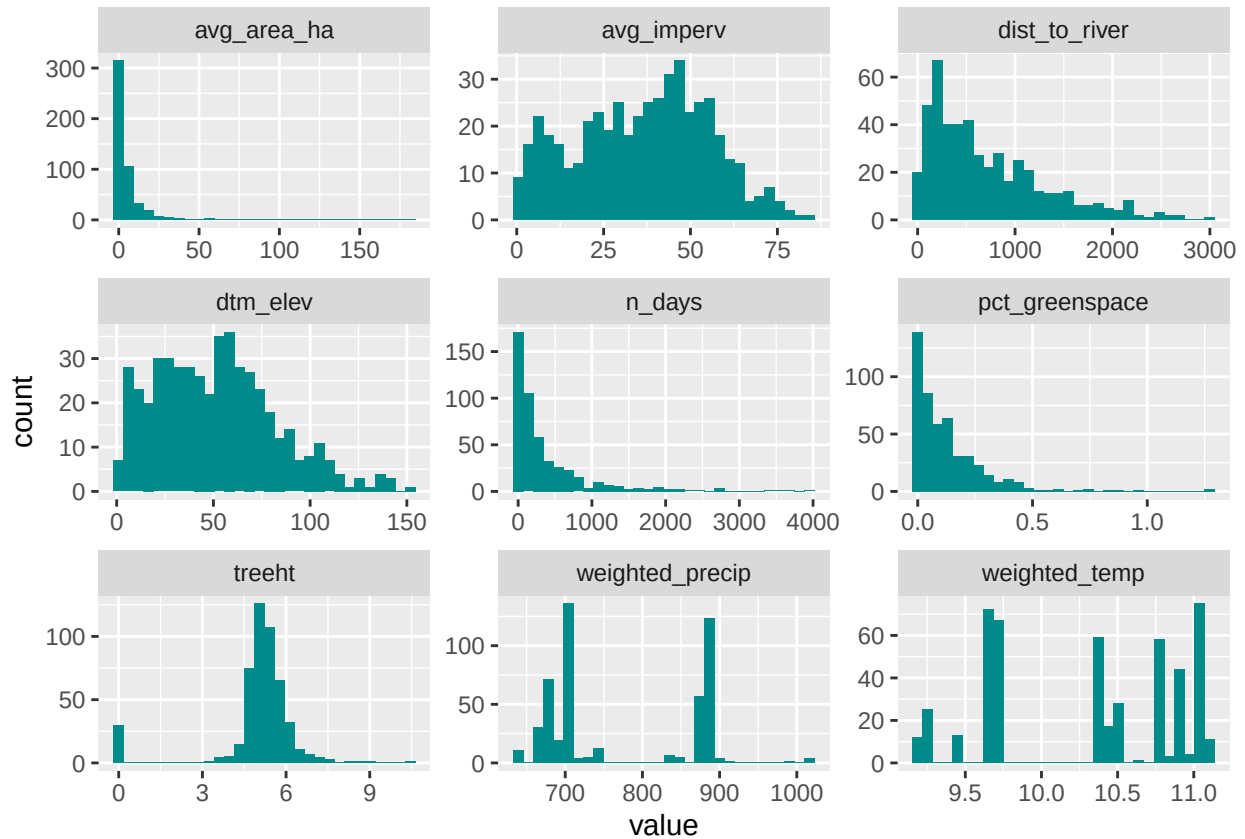
Distributions

The distribution of these numerical variables is varied. There is some skew in the data and multiple peaks in the the precipitation and temperature categories. This is to be expected since the cities are geographically distinct from each other.

```
dist_hists <- df %>%
  pivot_longer(all_of(numerical_predictors)) %>%
  ggplot(aes(value)) +
  geom_histogram(bins = 30, fill = "cyan4") +
  facet_wrap(~name, scales = "free")
```

```
dist_hists
```

```
## Warning: Removed 4 rows containing non-finite outside the scale range
## ('stat_bin()').
```

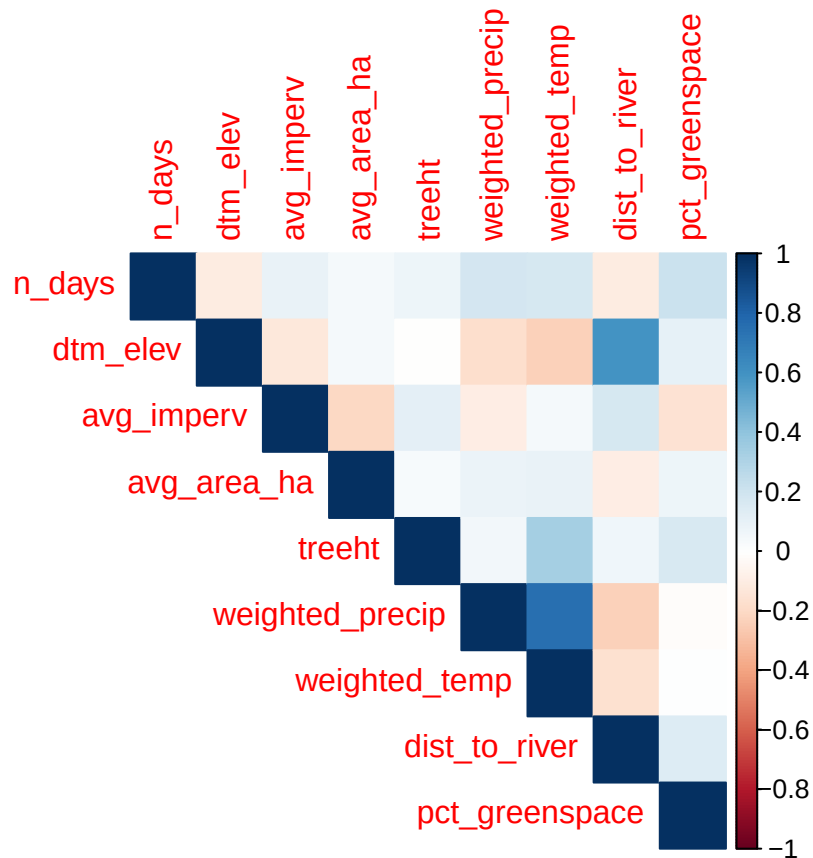


Correlation Matrix

Correlation analysis indicates that there is very little correlation between variables. The strongest correlations lie between distance to river and elevation, and precipitation and temperature. These are both environmentally logical.

```
library(corrplot)

corrplot(
  cor(df[numerical_predictors], use = "pairwise.complete.obs"),
  method = "color",
  type = "upper")
```



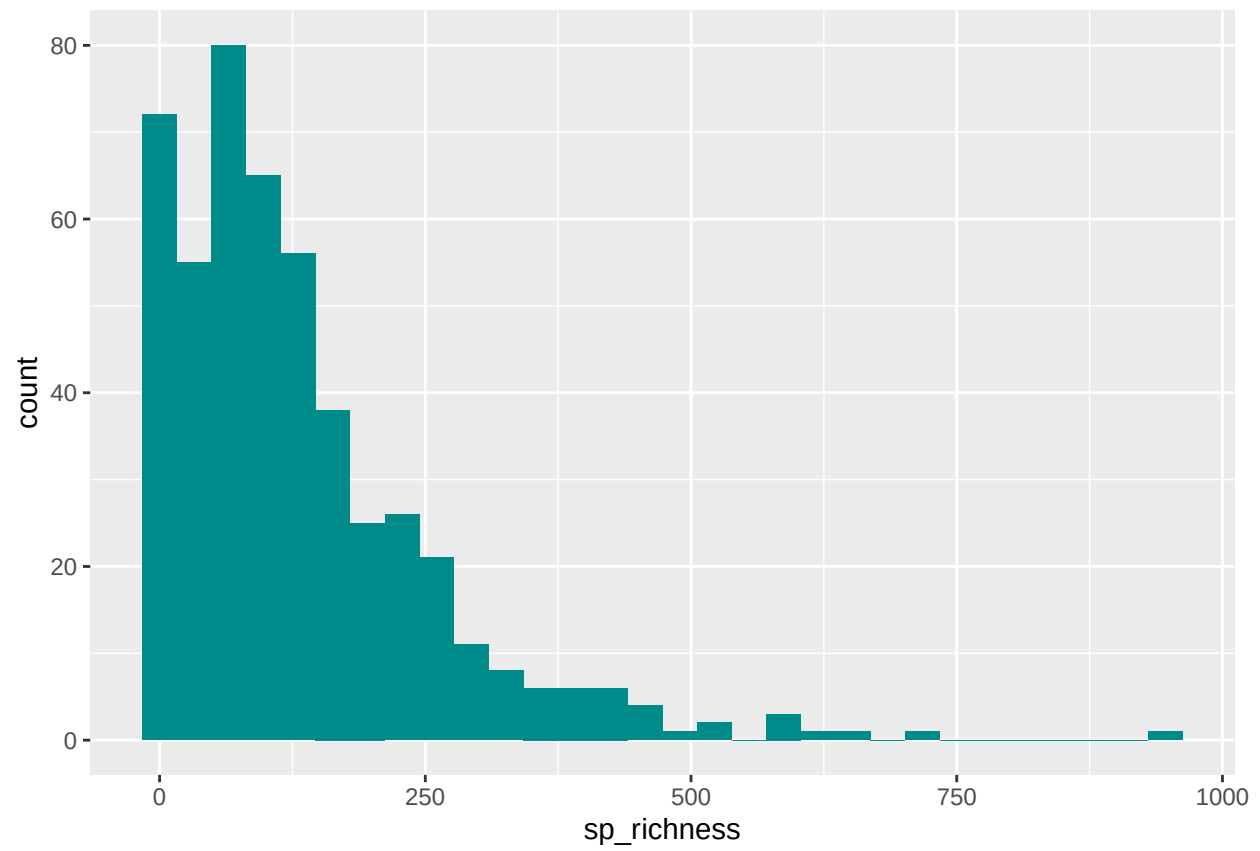
Richness and Simpson

These histograms show non-normally distributed data. The richness data is heavily skewed and overdispersed, with a few extreme outliers. The Simpson histogram also shows highly skewed data, bound between 0 and 1, but with a small, long tail. To account for this, a negative binomial regression will be used for the richness diversity indice and a beta regression model for the Simpson diversity indice.

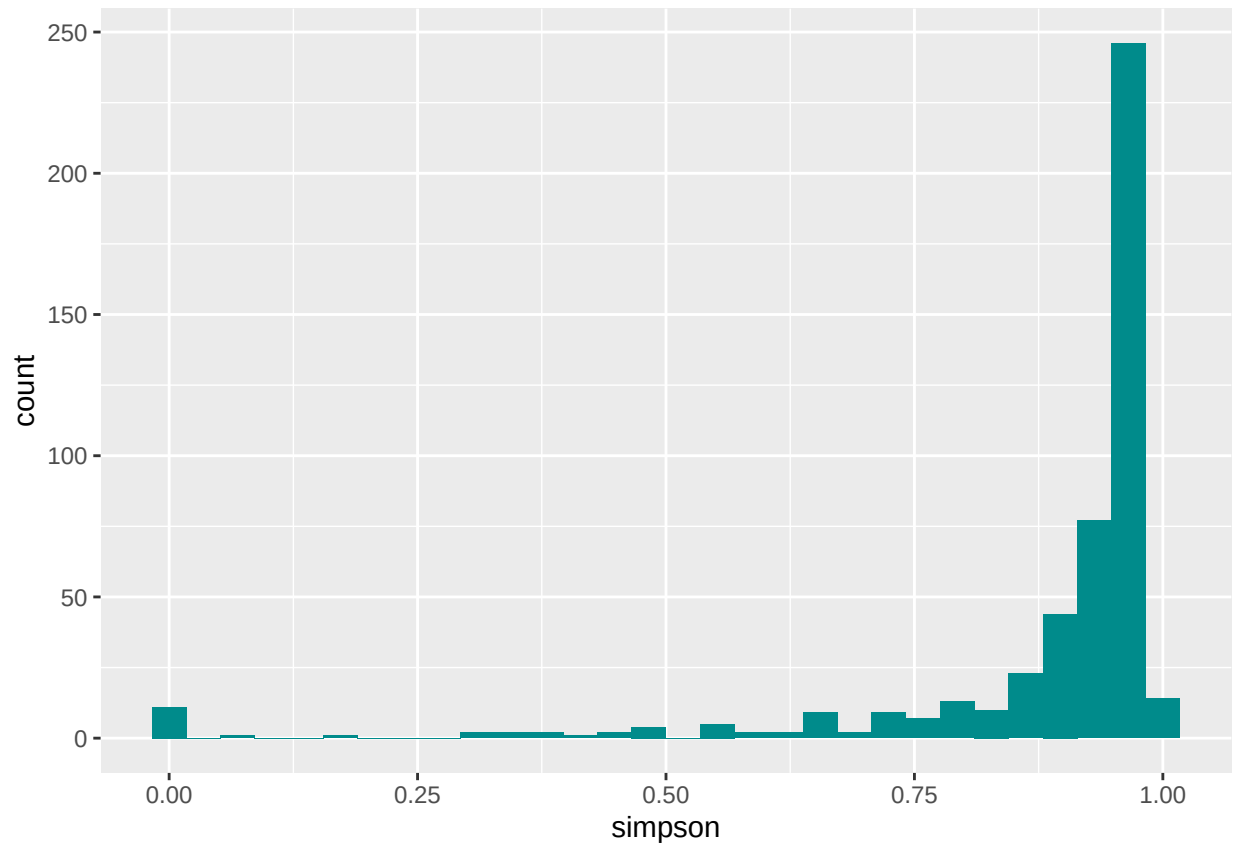
```
rich_hist <- ggplot(df, aes(sp_richness)) +
  geom_histogram(fill = "cyan4")

simp_hist <- ggplot(df, aes(simpson)) +
  geom_histogram(fill = "cyan4")

rich_hist
```



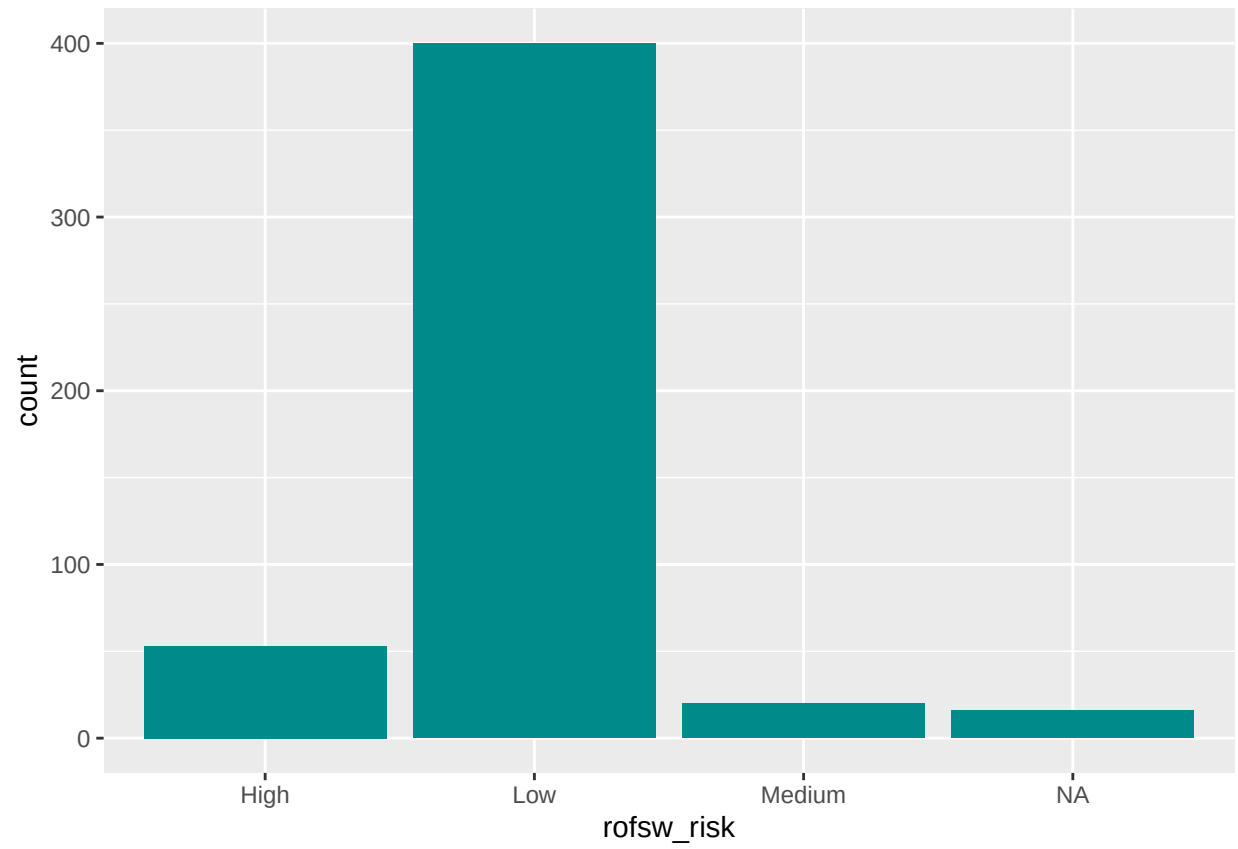
simp_hist



ROFSW and ROFRS

These bar plots show the spread of categorical variables, Risk of flooding from surface water (ROFSW) and Risk of flooding from rivers and seas (ROFRS). The variance between categories is large in both variables, but especially the ROFSW. In ROFRS, there is a huge number of grid cells with no data associated. This may prove an unreliable predictor when modelling biodiversity.

```
rofsw <- ggplot(df, aes(rofsw_risk)) +  
  geom_bar(fill = "cyan4")  
  
rofrs <- ggplot(df, aes(rofrs_risk)) +  
  geom_bar(fill = "cyan4")  
  
rofsw
```



rofrs

